Neuroendocrine tumors — gastroenteropancreatic

Managed on two axes: **differentiation** (well-differentiated NET vs poorly differentiated NEC) and **grade** (Ki-67 / mitotic count) · then modified by **primary site** and whether the tumor is **functional**

Stanford Oncology Handbook 2025, Neuroendocrine Tumors (GI section) · AJCC 8th · items marked † are outside the handbook chapter

WHO CLASSIFICATION

Grade	Differentiation	Ki-67	Mitoses /10 HPF
1 — low	Well-differentiated	0–<3%	<2
2 — intermediate	Well-differentiated	3–20%	2–20
3 — high	**Well-differentiated**	>20%	>20
3 — high	**Poorly differentiated**	>20%	>20

Grade is the higher of Ki-67 or mitotic count. Poorly differentiated = **neuroendocrine carcinoma (NEC)**; well differentiated = **NET**. **Well-differentiated G3 is a distinct sub-group** — well-differentiated histology with Ki-67 >20%, and its behavior varies widely. It is **not** the same as NEC. Mixed neuroendocrine + adenocarcinoma = **MANEC**. "Carcinoid" is the older term for NET.

GASTRIC NET TYPES

Type	Gastrin	%	Setting
1	↑	70–80%	Autoimmune **atrophic gastritis**; elderly, female, pernicious anemia
2	↑	5–8%	Gastrin-secreting tumor — **MEN1, Zollinger-Ellison**
3	normal	20%	Solitary nodule >1.5 cm, **healthy gastric mucosa**; usually symptomatic

Type applies to **gastric NETs only**. Check gastrin at baseline and on follow-up for gastric primaries.

SITE-SPECIFIC POINTS

- Frequency: **GI tract > lung > other**. Most are sporadic; a minority arise in **MEN1**/**MEN2**
- **Small bowel:** the classic carcinoid-syndrome primary (with lung); **consider resecting the mesenteric mass prophylactically** to avoid future complications
- **Pancreatic:** broadest baseline hormone panel; **sunitinib is approved here only**; CAPTEM and temozolomide data are in pancreatic G1–G2
- **Carcinoid syndrome:** small bowel and lung primaries with elevated serotonin — flushing is the archetypal functional symptom. If the tumor is functional, somatostatin analogue therapy is **indefinite**
- † Telotristat and carcinoid crisis are not addressed in this chapter
- Staging follows AJCC 8th by primary site (stomach, small bowel/ampulla, pancreas, colon/rectum)

TREATMENT PATHWAY

Localized / resectable

↓

Resect the primary if possible
there is no role for adjuvant therapy in NETs
small bowel: also resect the mesenteric mass

↓

Low-burden, unresectable, non-functional
watch and wait is acceptable

Advanced well-differentiated NET (G1–G2)

↓

Start with a somatostatin analogue
octreotide LAR 30 mg IM or lanreotide 120 mg SQ every 4 weeks — PROMID, CLARINET
antitumor *and* hormonal control

↓

Then by site and burden
PRRT Lu-177 dotatate (NETTER-1)
everolimus any site · **sunitinib pancreatic only**
cabozantinib pancreatic and extrapancreatic

↓

Pancreatic G1–G2
CAPTEM (PFS 23.2 mo) or temozolomide (15.1 mo)
cap at ~12 months

↓

Liver-dominant disease
chemoembolization · radioembolization
bland embolization · microwave ablation

Ki-67 >20% — well-diff G3 or NEC

↓

Treat as an aggressive tumor
FOLFOX or CAPOX
cisplatin/etoposide or carboplatin/etoposide
ipilimumab + nivolumab is an option

↓

But G3 NET is not NEC
well-differentiated G3 behaves variably
NETTER-2 enrolled high-risk **G2–G3** GEP-NETs

REGIMENS & DOSING

Agent / regimen	Dosing	Schedule
Octreotide LAR	30 mg IM	q4 weeks
Lanreotide	120 mg deep subcutaneous	q4 weeks
Lu-177 dotatate (Lutathera)	7.4 GBq (200 mCi) IV	q8 weeks × **4 doses**
Everolimus	10 mg PO daily	continuous
Sunitinib	37.5 mg PO daily — **pancreatic NET only**	continuous
Cabozantinib	60 mg PO daily	continuous
CAPTEM	capecitabine 750 mg/m² PO BID d1–14 + temozolomide 200 mg/m² PO d10–14	q28d, **≤12 months**
Temozolomide alone	200 mg/m² PO d1–5	q28d, ≤12 months
FOLFOX	5-FU 1200 mg/m²/day CIV 48 h + oxaliplatin 85 mg/m² d1	q14d
CAPOX	capecitabine 850 mg/m² PO BID d1–14 + oxaliplatin 130 mg/m² d1	q21d
Cisplatin / etoposide	cisplatin 60 mg/m² d1 + etoposide 100 mg/m² d1–3	q21d
Carboplatin / etoposide	carboplatin AUC 5 d1 + etoposide 100 mg/m² d1–3	q21d
Ipilimumab + nivolumab	ipilimumab 1 mg/kg q6wk + nivolumab 240 mg q2wk	as above

Pre-medicate temozolomide with **ondansetron 8 mg PO 30 minutes prior**. Temozolomide-containing regimens are typically capped at ~12 months.

WORKUP — IMAGING & GENETICS

- Multiphasic CT or MRI; for liver metastases, **MRI with Eovist** may help
- **Ga-68 DOTATATE PET at baseline, then every 2 years**, or for a specific clinical question — **not a routine restaging scan**
- Endoscopy as indicated
- **Brain MRI** only for poorly differentiated disease with symptoms or raised suspicion
- **MR spine** for symptoms, or considered annually for axial bone metastases
- **Genetics referral** if diagnosed **<50 years**, family history suggesting **MEN1**/**MEN2**, or a somatic NGS finding with germline implications (BRCA1/2, MEN1, Lynch genes)

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BIOCHEMICAL WORKUP BY SITE

Primary	Baseline / pre-op	Follow-up
Pancreatic	Chromogranin A · gastrin · glucagon · insulin/proinsulin · NSE · pancreatic polypeptide · VIP · 24-h urine **5-HIAA**. As indicated: PTHrP, ACTH, GHRH	Chromogranin A, NSE, and **only those elevated at baseline**
Small bowel / other	Chromogranin A · NSE · 24-h urine **5-HIAA** · gastrin if gastric primary	Same panel

All serum labs preferably **fasting**; metastatic follow-up interval is usually **q1–3 months** as clinically indicated. **Follow only the markers that were elevated at baseline.**

> **24-hour urine 5-HIAA requires a 3-day serotonin-restricted diet**, before and during collection. Avoid: **bananas, plums, pineapple, kiwi, avocado, eggplant, tomato products, nuts (especially walnuts), coffee, tea and chocolate**.
>
> Medications to avoid where feasible: cough and antihistamine preparations, antihypertensives, fluorouracil, MAO inhibitors, acetaminophen, aspirin, melphalan — **weigh risks and benefits before stopping anything essential**.

MONITORING ON THERAPY

- **On somatostatin analogues:** check **vitamin B12, vitamin D and TSH at least annually**
- SSA adverse effects: injection-site pain, abdominal pain, bloating, diarrhea, **gallstones**, hyper- or hypoglycemia, pancreatic exocrine insufficiency
- **Everolimus:** stomatitis, rash, fatigue, diarrhea, infection, elevated lipids, **pneumonitis**
- **Sunitinib:** diarrhea, nausea, fatigue, stomatitis, rash
- **Cabozantinib:** hypertension, fatigue, diarrhea, hand-foot syndrome, **thromboembolic events**

TRIAL EVIDENCE

PROMID	octreotide — TTP **14.3 vs 6 mo** vs placebo (Rinke, JCO 2009)
CLARINET	lanreotide — PFS **38.5 mo** (Caplin, NEJM 2014)
NETTER-1	Lu-177 dotatate, 2nd line+, G1/G2 — PFS not reached vs 8.5 mo; 2018 update median OS **NR vs 27.4 mo** (Strosberg, NEJM 2017)
NETTER-2	Lu-177 dotatate **first line**, high-risk SSTR-positive G2–G3 GEP-NET — PFS **22.8 vs 8.5 mo** vs high-dose octreotide. **Not yet FDA approved** (Singh, Lancet 2024)
COMPETE	Lu-177 edotreotide vs everolimus, G1/2 GEP-NET — PFS **23.9 vs 14.1 mo**; OS trending favorable, not yet significant
CABINET	cabozantinib — extrapancreatic PFS **8.4 vs 3.9 mo**; pancreatic **13.8 vs 4.4 mo** (Chan, NEJM 2024)
RADIANT-3	everolimus, pancreatic — PFS **11.0 vs 4.6 mo** (Yao, NEJM 2011)
RADIANT-4	everolimus, non-functional GI/lung — PFS **11.0 vs 3.9 mo** (Yao, Lancet 2015)
Sunitinib	pancreatic NET only — PFS **11.4 vs 5.5 mo** (Raymond, NEJM 2011)
CAPTEM (E2211)	pancreatic G1/G2 — PFS **23.2 mo** vs temozolomide alone **15.1 mo** (Kunz, JCO 2022)

PEARLS

- **Grade is the higher of Ki-67 or mitotic count** — not an average, and not Ki-67 alone
- **Well-differentiated G3 is not NEC.** Same Ki-67 threshold, different disease, and behavior varies widely
- **There is no role for adjuvant therapy in NETs** — resect and observe
- **Sunitinib is pancreatic-only.** CAPTEM and temozolomide data are also pancreatic G1–G2. Cabozantinib is the one with both pancreatic and extrapancreatic data
- **Ga-68 DOTATATE PET is not a routine restaging scan** — baseline, q2 years, or a specific question
- **Follow only the tumor markers that were elevated at baseline**
- **NETTER-2 (first-line PRRT in G2–G3) is not FDA approved yet**; COMPETE's OS is not yet significant
- † The chapter is GEP-NET focused — **lung/bronchial carcinoid, thymic NET and pheochromocytoma/paraganglioma are separate entities not covered here**